Neurocomputing 311 (2018) 99–111

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

A novel framework of continuous human-activity recognition using Kinect

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ARTICLE INFO

Article history:
Received 12 October 2016
Revised 9 May 2018
Accepted 15 May 2018
Available online 25 May 2018

Communicated by Dr. Zhang Zhaoxiang

Keywords:
Continuous activity
Depth sensors
HMM
BLSTM-NN
Kinect

ABSTRACT

Automatic human activity recognition is being studied widely by researchers for various applications. However, majority of the existing work are limited to recognition of isolated activities, though human activities are inherently continuous in nature with spatial and temporal transitions between various segments. Therefore, there are scopes to develop a robust and continuous Human Activity Recognition (HAR) system. In this paper, we present a novel Coarse-to-Fine framework for continuous HAR using Microsoft Kinect. The activity sequences are captured in the form of 3D skeleton trajectories consisting of 3D positions of 20 joints estimated from the depth data. The recorded sequences are first coarsely grouped into two activity sequences performed during sitting and standing. Next, the activities present in the segmented sequences are recognized into fine-level activities. Activity classification in both stages are performed using Bidirectional Long Short-Term Memory Neural Network (BLSTM-NN) classifier. A total of 1110 continuous activity sequences have been recorded using a combination of 24 isolated human activities. Recognition rates of 68.9% and 64.45% have been recorded using BLSTM-NN classifier when tested using length-modeling and without length-modeling, respectively. We have also computed results for isolated activity recognition performance. Finally, the performance has been compared with existing approaches.

1. Introduction

Over last two to three decades, Human Activity Recognition (HAR) has emerged as an important research area in Human Computer Interaction (HCI). The goals of a HAR system consist of modeling of user's behavior and assisting them by automatic identification of their tasks. A large number of applications are directly based on HAR system including automatic video surveillance systems, pedestrian detecti on method or healthcare automation systems. However, there are several challenges associated with HAR systems, namely variety in activities, complex human motions, low inter-class similarities, and high intra-class similarities between different activities [26]. Majority of the existing approaches are based on extracting spatio-temporal information from RGB video frames for recognition of activities [20,27]. It includes recognition of human gestures and activities in 2D images or video frames. In real life applications, activities usually occur contiguously. However, a large segment of these applications focus on recognition of isolated activities. To make a HAR system to be applicable to real world scenarios, there is a need to implement typical continuous HAR systems. One example of a real-life scenario is depicted in Fig. 1(a), where a person continuously performs activities, like *sitting, drinking*, $phone - call$, *standing*, and *boxing*. These activities are continuously recorded in a video sequence. Therefore, to recognize these activities, an efficient and accurate segmentation is required before recognition. In this work, we propose a continuous HAR framework which performs en embedded segmentation and recognition together. A coarse-level grouping of similar activities that are done during sitting and standing and their sequence alignment are performed before segmentation of individual actions. The scenario of sequence grouping is shown in Fig. 1(b), where the activities are aligned into two groups i.e., *sitting, drinking*, and $phone - call$ as "$Group - Sitting$" and *standing* and *boxing* as "$Group - Standing$". Next, the activities present in each coarse-level sequence are recognized at a finer level using sequence modeling classifier as shown in Fig. 1(c).

With the advancement of depth sensor technology such as Microsoft Kinect, it became possible to capture both color and depth images simultaneously. The depth images can provide accurate data about shape and geometrical structure and they have several advantages over RGB images in terms of background segmentation,

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https://doi.org/10.1016/j.neucom.2018.05.042
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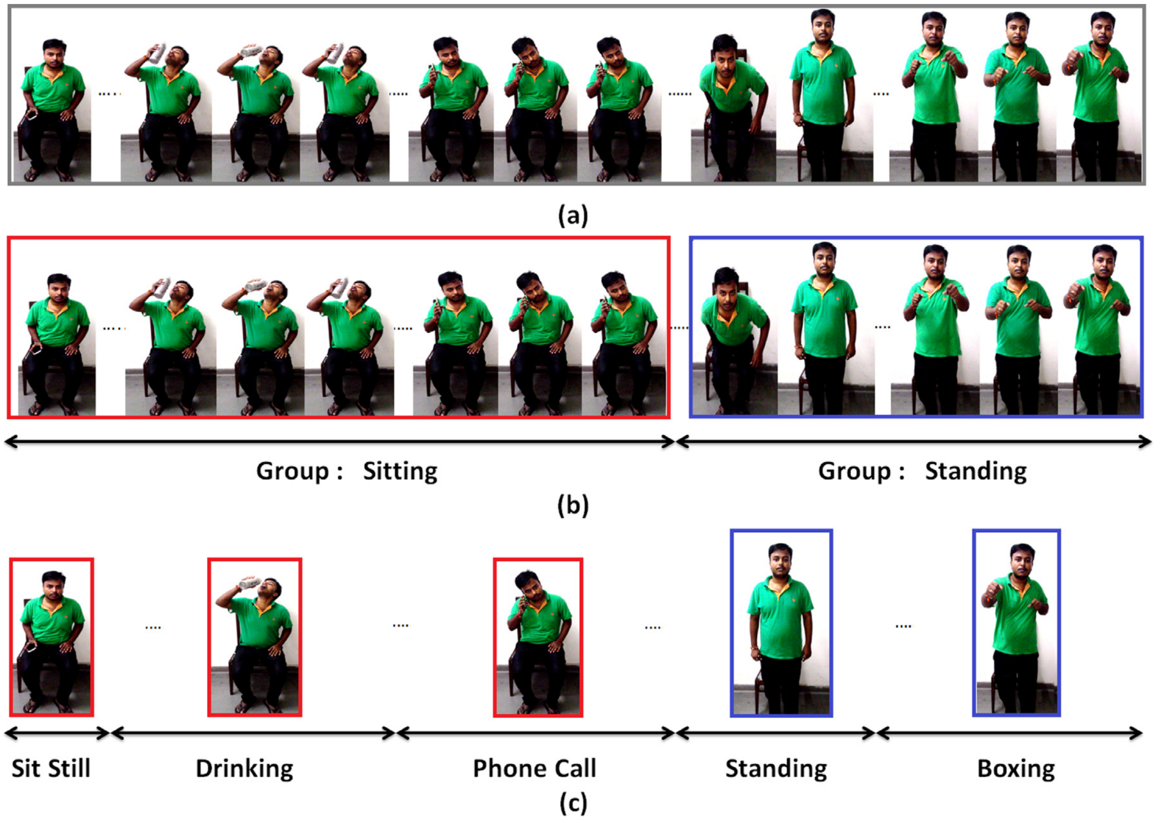

Fig. 1. Example of continuous activity sequence (a) A user performs five activities continuously (b) Grouping of activities during sitting and standing (c) Individual recognition of grouped activities.

object detection, and activity recognition. Jalal et al. [14] have proposed a HAR system using depth video that works well within indoor environments. The authors have shown that the activity recognition using depth-map performs better than conventional 2D image based recognition methods. Thus, there has been a notable progress [4–6,36] in developing HAR systems using depth information. Using a single depth image, estimation of positions of various body joints in the form of skeleton or body structure, is now possible. Devanne et al. [8] have utilized the 3D skeleton points to develop a HAR system using Kinect. The 3D activity sequences are represented as spatio-temporal motion trajectories of pose vectors in Euclidean space followed by an open nonlinear modeling of the temporal variations.

Activity recognition using accelerometer and gyroscopes have also been investigated by researchers [34]. Such systems can facilitate assistance to the users. Yan et al. [39] have proposed a continuous HAR system for measuring daily life activities in mobile phones using embedded accelerometers. They have presented an activity-sensitive approach using frequency as feature in real time classification. Bidirectional Long Short-Term Memory Neural Network (BLSTM-NN) is a popular sequence classification model that has been successfully used in various hand-writing and gesture recognition problems [17,19]. The classifier can model long term dependencies without requiring pre-segmentation of sequences. Graves et al. [10] have used BLSTM-NN to develop a handwriting recognition system. The authors have shown that the word recognition performance using BLSTM-NN outperformed other sequential classifiers such as HMM. Similarly, the authors in [19] have used BLSTM for the recognition of text written in air (3D space) using Leap motion sensor.

In this work, we present a framework for continuous HAR system using Kinect. The activities are represented as 3D trajectories of twenty body joints of human torso. Our proposed framework does a coarse-level segmentation between activities performed while standing and sitting. Next, segmented sequences are recognized at a finer level using sequence classifiers. The main contributions of the paper are as follows:

1. Firstly, we present a coarse-to-fine framework to capture continuous human activities to perform grouping of activities during sitting and standing (stage 1) and individual activities (stage 2).
2. Secondly, recognition of the segmented sequence has been performed using BLSTM-NN. A comparative performance analysis with full sequence recognition has also been presented with BLSTM-NN and HMM classifiers.

Rest of the paper is organized as follows. A survey of the research work in this area, is discussed in Section 2. Section 3 describes our proposed methodology of continuous-HAR system followed by feature extraction, and recognition. Results are presented in Section 4. Finally, we conclude with some future possibilities of the work in Section 5.

2. Related work

Here, we summarize the details of activity recognition systems proposed by various researchers using 2D RGB images [29], motion sensors [34], and RGB-D data streams [38]. The approaches are mainly divided into two categories. Some researchers focus on the recognition of isolated activities [31,42] and others concentrate on the recognition of continuous activities [41].

2.1. Isolated activity recognition using Kinect

There exist computer vision-based research work that perform isolated action/activity recognition in video [1,15,37]. However, the recognition performance of such systems in real-world scenario usually vary significantly. Kinect offers a low-cost and real-time solution for the 3D estimation of the objects and persons appeared in its viewing field. The issues of capturing RGB as well as depth streams can be resolved using such systems. Researchers have successfully used the device to build activity recognition systems [28,38]. Gaglio et al. [9] have proposed a HAR system using Kinect by tracking 3D joints of human torso. Detection and classification of body postures are performed using k-mean and SVM classifiers, respectively and recognition is performed using HMM. In [28], the authors have presented an action recognition system using Kinect. Feature vector has been built using 3D joint orientation of human parts. DTW has been applied for recognition of five actions. However, their system fails to recognize complex human actions when performed by multiple people. A prototype system for object detection, tracking and recognition is presented in [26]. It uses a combination of color and depth data acquired using Kinect. Instead of tracking and recognition of single human motions, the system is able to recognize human actions and kitchen objects simultaneously. In [31], the authors have proposed a Local Occupancy Patterns (LOP) to capture local occupancy features at joint locations to discriminate different types of interactions with objects. The 3D point cloud generated at each joint are partitioned into a spatial grid. The occupancy features are obtained by counting and normalizing the number of points that fall into each bin of the grid. The approach has been tested on two different 3D action datasets with recognition rates of 88.2% and 87.75%, respectively. A kinect based virtual trainer system for exercise feedback has been proposed in [22]. The system uses kinect skeleton to analyze how accurate a person perform an exercise.

A new descriptor for activity recognition systems using depth has been proposed in [21]. A depth sequence is described using a histogram that captures the distribution of surface-normal orientation in 4D space including time, depth, and spatial coordinates. The approach has been tested on three 3D activity datasets with accuracies of 88.89%, 92.45%, and 96.67%, respectively. The authors of [40] have clustered the hyper-surface normal to form polynomials from a depth sequence and used them to characterize the local motion and surface information for activity recognition. The spatial and temporal orders are captured using a spatio-temporal pyramid that subdivide the depth sequence into a set space-time grids. Next, aggregation the low-level polynormals into a super normal vector has been done. The approach has been tested on multiple datasets.

There exists works for activity recognition based on RGB-Depth, and skeleton [13,32,35]. Hu et al. [13] have proposed activity recognition system using heterogeneous features from RGB and depth information. Their model was capable of learning a set of subspaces making features with distinct dimensions using intermediate transforms. The model aims at joint learning by feature subspaces based on individual RGB and Depth channels and their shared subspaces. Likewise, an actionlet ensemble learning based framework has been proposed in [32] for 3D human action recognition. Their framework models the interactions of subsets of human joints for capturing human motion and human-object interactions. The LOP based feature has been used for representing depth appearance in the neighborhood. The temporal structure has been represented using Fourier Temporal Pyramids (FTP) based feature. In [35], a human–object interaction framework for both event and object recognition together in 4D space has been proposed. The authors have designed dynamic programing based beam search algorithm for video segmentation, event recognition, and object detection simultaneously. The framework has been tested on a RGBD dataset of 3815 videos with an accuracy of 87% in event recognition.

2.2. Continuous activity recognition approaches

In this section, we provide the details of continuous-HAR system proposed by various researchers using 2D RGB images [29], motion sensors [34] and RGB-D data streams [38]. Kulkarni et al. [16] have proposed a continuous activity recognition system in videos using dynamic programming. The authors have utilized Dynamic Time Warping (DTW) algorithm for the alignment of test sequences and the recognition of activities has been performed on per-frame representation of videos. As humans perform similar activities with different speeds, it usually causes large variation within the same activity. A speed invariant HAR system has been proposed in [7]. The system is able to work even when some portion of the data is missed due to occlusion or noises. They have used Fourier Transformation and Bayesian Network for recognition of two activities performed by five users. In [30], the authors have used human posture features using spatial and temporal variations applied on input image sequences. A Gaussian model has been used for blob detection. Tracking has been done using Kalman filter. A fusion of HMM is also considered that treats each feature independently and decide the final activity recognition. Recently, the authors of [12] have used deep learning concept for developing a continuous activity learning framework. The features of segmented activities from the streaming videos are learned using a sparse auto-encoder. Bloom et al. [3] have proposed a framework for action recognition applicable to real-time gaming using Kinect. The authors have extracted a 170 dimension feature vector from 3D skeleton that includes pairwise joint position difference, joint velocity, velocity magnitude, joint angle velocity in xy and xz planes, and 3D joint angles between three distinct joints. The action-based and frame-based F1 scores of fighting sequences are reported as 46.66% and 61.17%, respectively.

Accelerometer, Global Positioning System (GPS), or microphone have also been used for developing HAR systems. Ward et al. [34] have proposed a continuous HAR system using body-attached sensors, namely accelerometer and microphone. Activity segmentation from the continuous data streams can be performed by analyzing the sound intensity. Recognition of segmented activities has been performed using linear discriminant analysis (LDA) and HMM on audio and acceleration data with recall and precision of 78% and 74%, respectively. In [33], a continuous auto-encoder based HAR system has been presented using wearable sensors. The authors have extracted time-frequency domain features from the sensor data and used Deep Belief Network (DBN) for recognition of activities with identification rate of 99.3%. A summary of the existing work is presented in Table 1 for providing more details to the reader.

3. Proposed system

In this section, we present a detailed description of the proposed framework of continuous-HAR. In order to capture 3D continuous human activities, we have used Kinect Xbox 360 [43]. The device comes with its Software Development Kit (SDK) that provides 3D point cloud of observed scene occurred in the viewing field of the sensor. The device has a depth sensing range between 800 to 4000 mm. Using the depth sensor, the device provides a skeletal view of human joints consisting of twenty 3D points. The skeletal representation of the human torso has been used in this work. The complete flow-diagram of our approach is depicted in Fig. 2, where captured 3D trajectories are further processed for

Table 1
Summary of the related work.

Author name & year	Methodology (Features & classifier)	Dataset	Accuracy
Isolated activity recognition approaches			
Gaglio et al. [9], 2015	3D joints, k-means SVM and HMM	18 activities	84.8% precision & 84.5% recall
Sempena et al. [28], 2011	Joints orientation, DTW	5 activities	N.a.
Rohrbach et al. [26], 2012	Body model, joint velocity FFT features	65 cooking activities	45.1%
Wang et al. [31], 2012	3D joints, LOP	20 actions	88.2%
Oreifej et al. [21], 2013	Histogram of 4D normals, surface normal vector	20 actions, 12 gestures, 6 action pairs	88.89%, 92.45% 96.67%
Yang et al. [40], 2014	Hypersurface normals, super normal vectors	20 actions, 12 gestures, 6 action pairs, 16 daily activities	93.09%, 94.72%, 98.89%, 86.25%
Wang et al. [32], 2014	Actionlet ensembling, LOP	8 activities	88.34%
Wei et al. [35], 2013	Beam search, expectation maximization, motion template, HMM	8 events, 11 objects	87.0%
Hu et al. [13], 2017	I-transform, heterogeneous feature fusion	12 activities	84.89%
Continuous activity recognition approaches			
Kulkarni et al. [16], 2015	2D video, DTW	8 actions	45.7%
Len et al. [7], 2002	Fourier transformation, Bayesian networks	2 activities	90%
Tran et al. [30], 2010	Posture features, Kalman filter, HMM	5 activities	99.22%
Hasan et al. [12], 2015	Video stream, deep features, autoencoder	multiple datasets	–
Bloom et al. [3], 2012	3D joint position, velocity, magnitude, angle velocity	20 gaming activities	61.17% F1-score
Ward et al. [34], 2006	Accelerometer, microphone sound intensity, LDA, HMM	21 activities	78% recall, 74% precison
Wang et al. [33], 2016	PCA, autoencoder, time-frequency features, DBN	19 activities	99.3%

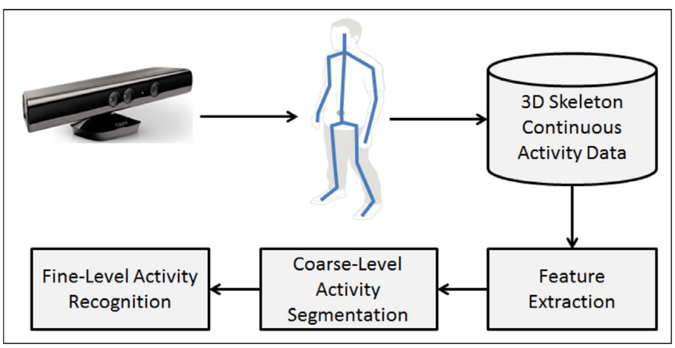

Fig. 2. Flow-diagram of the proposed framework of continuous activity recognition.

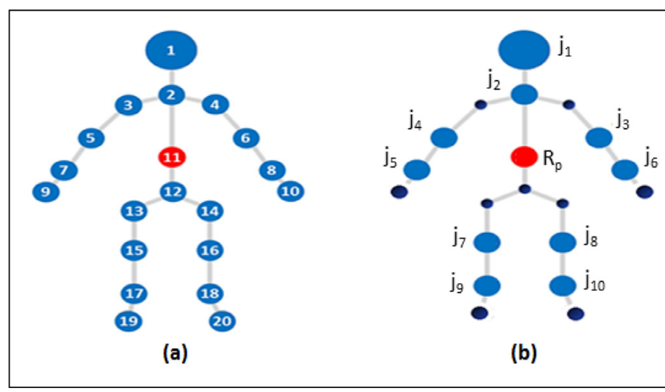

Fig. 3. 3D joint selection as per the evolutionary algorithm with a reference point R_p. J denotes positions of 10 body joints [9].

feature extraction and sequence alignment before recognition of an activity.

3.1. Feature extraction

This step is used to extract an appropriate set of observable features from the recorded data that helps to improve the recognition rate. In this work, we have constructed a feature vector (F_T) made by combining five features including 3D joints position (F_j), two angular features (A_v) & (A_d), distance (F_d), and velocity (F_v). The feature vector (F_T) is described in (1).

$$F_T = \{F_j, A_v, A_d, F_d, F_v\} \quad (1)$$

3.1.1. 3D joints position

Using the device's Application Programming Interface (API), we have recorded 3D positions of twenty body joints as depicted in Fig. 3(a). However, every joint may not be highly informative due to various types of noises present in the device and structure of the human body. Therefore, only relevant joints locations have been identified and used through an evolutionary algorithm that computes an optimal set of joint positions. For this, we have adopted the approach presented in [9] that removes redundant joints due to their closeness and are not relevant for activity recognition. Using this, we have extracted a feature vector (F_j) consisting 3D positions of ten joints as depicted in Fig. 3(b). F_j is presented in (2). In addition to this, we have also extracted one additional joint that corresponds to '*Spine*'. This point acts as a reference point for further feature computation and it is denoted by R_p as depicted in Fig. 3(b),

$$F_j = \{J_1, J_2, \ldots, J_{10}\} \quad (2)$$

where $J_1, J_2, \ldots, J_{10}$ denote positions of the selected body joints.

3.1.2. Angular features

Using F_j, we have extracted two angular features by computing the angular features (A_v) of the body points with respect to the reference R_p. The direction angles are computed using the joint sequence (A_d) as follows:

(i) *Angular features using reference point (A_v):* Computation of A_v is done in each frame f_i of an activity sequence, say $f_1, f_2, \ldots, f_n$ of n with respect to R_p. Using this, ten different vectors, e.g. $\overrightarrow{RP_1}, \overrightarrow{RP_2}, \ldots, \overrightarrow{RP_{10}}$ are computed using R_p. Next, the angles with the coordinate axes are computed using these vectors.

(ii) *Direction based angular feature (A_d):* Computation of A_d is done for each joint sequence present in F_j. For a given 3D trajectory that represents a joint, direction of the point $Q(l, m, n)$ is estimated using its two neighbor points, i.e. $P(l_1, m_1, n_1)$ and $R(l_2, m_2, n_2)$. Next, a vector $\overrightarrow{PR}$ is computed that makes θ_x, θ_y and θ_z angles with the coordinate

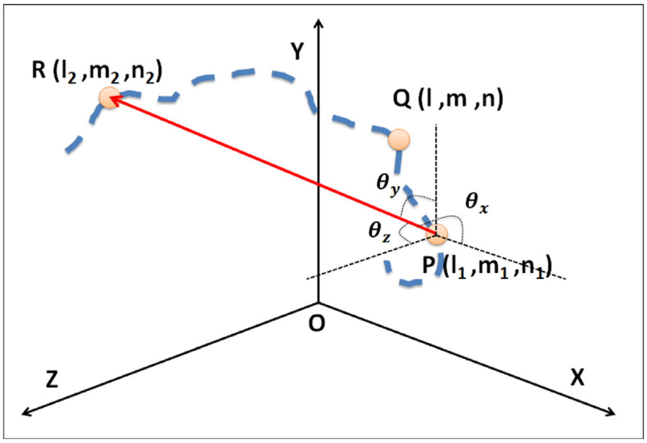

Fig. 4. Estimation of the angular directional features using neighbor points.

axes as depicted in Fig. 4. The angles can be calculated using (3) and (4). We have computed these angles and are considered as direction-based angular features.

$$\vec{s} = \vec{PR} = < s_l, s_m, s_n >, |\vec{s}| = \sqrt{s_l^2 + s_m^2 + s_n^2} \tag{3}$$

$$cos(\theta_x) = \frac{s_l}{|\vec{s}|}, cos(\theta_y) = \frac{s_m}{|\vec{s}|}, cos(\theta_z) = \frac{s_n}{|\vec{s}|} \tag{4}$$

3.1.3. Distance feature (F_d)

Using F_j, computation of F_d features has been done by finding the distance of joints from the reference point F_p. The computation is done at each frame using (5), where (x_r, y_r, z_r) denotes the coordinates of the reference point sequence, (*x*, *y*, *z*) represents the 3D coordinate corresponding to a joint, and *D* is the distance.

$$D = \sqrt{(x_r^2 - x)^2 + (y_r^2 - y)^2 + (z_r^2 - z)^2} \tag{5}$$

Using this, we compute ten one-dimensional distance features that correspond to our feature vector D_d.

3.1.4. Velocity features (F_v)

A large number of variations can be seen in the activities as different activities are performed at varying speeds. For example, an ac tivity "**hand waving**" is performed at slow speed, whereas the activity "**toss paper**" is usually performed with higher velocity. Hence velocity feature computed from the distance between two successive points of the trajectory, such as (x_i, y_i, z_i) and ($x_{i+1}, y_{i+1}, z_{i+1}$) has been included. We have computed the velocity feature for each joint sequence present in the feature vector F_j that results into a new 30-dimensional feature vector corresponding to F_v.

3.2. BLSTM-NN based continuous activity recognition

BLSTM-NN is a popular sequence classification model and has been used in various hand-writing [10] and gesture recognition problems [17]. BLSTM-NN has two hidden layers that process a given input sequence in both directions [11]. One layer is used to process the sequence in forward direction, whereas the other processes the input in backward direction. Both hidden layers are attached with the same output layer that has one node for each possible activity present in the input sequence. A special node ϵ in the output layer is used to indicate '$no - activity$' which refers that no decision has been made at that position. The Connectionist Temporal Classification (CTC) objective function (*O*) is used as negative *log* probability of correct labeling of the entire training set. If *S* is the given training set with a pair of input and target sequence as (*p*, *q*), then the *O* can be described using (6)

$$O = -\ln\left(\prod_{(p,q)\epsilon S} p(q|p)\right) = -\sum_{(p,q)\epsilon S} \ln(p(q|p)). \tag{6}$$

Here, *O* has the ability to model a label sequence with given inputs. In some cases, it has been found that BLSTM-NN performs better than HMM classifier for handwriting recognition [10,11] by providing a long range of context accessible in both input directions. The model also resolves the problem of vanishing gradient using Long Short-Term Memory (LSTM) hidden layers which contains memory blocks to provide access to the information for longer time. In this work, the classifier has been learned with the feature vectors F_T for the recognition of continuous activities. In order to obtain best recognition results, BLSTM-NN has been trained multiple times and initialized randomly with different random seed for each initialization.

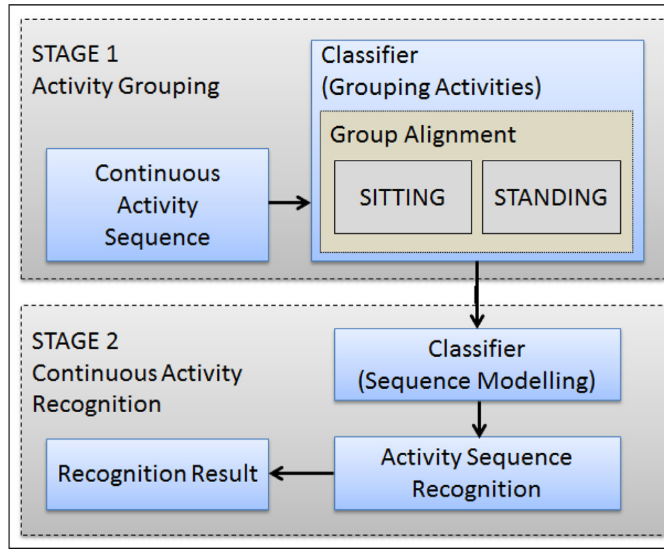

Fig. 5. Proposed model of two stage approach for continuous activity recognition.

3.3. Two-stage approach for continuous activity recognition

In this section, we discuss the proposed two stage *Coarse-to-Fine* approach for continuous activity recognition. Due to the presence of large vocabulary of actions, the recognition of continuous activity is a challenging task. To overcome this, the activity vocabulary may be divided into groups according to their similar nature of action and next recognition can be performed. With smaller number of actions in the vocabulary of each group, the recognition performance may improve.

With this idea, we have divided our activity recognition system in two stages. In stage 1, segmentation of continuous activity is performed broadly into two different groups in a coarser level, namely activities performed while sitting and standing. In stage 2, coarse-level activities are further segmented into finer level activities. A detailed block diagram of the scheme is depicted in Fig. 5, where an input sequence is grouped into two activities performed during sitting and standing using sequence alignment classifier. Next, the segmented sequences are passed to the second stage, where the sequences are modeled using a sequential classifier for recognition. The outcomes of stage 1 and stage 2 are shown in Figs. 6 and 7, respectively.

The recognition process of stage 1 and 2 can be carried out with a sequence modeling classifiers, such as Hidden Markov Mode,

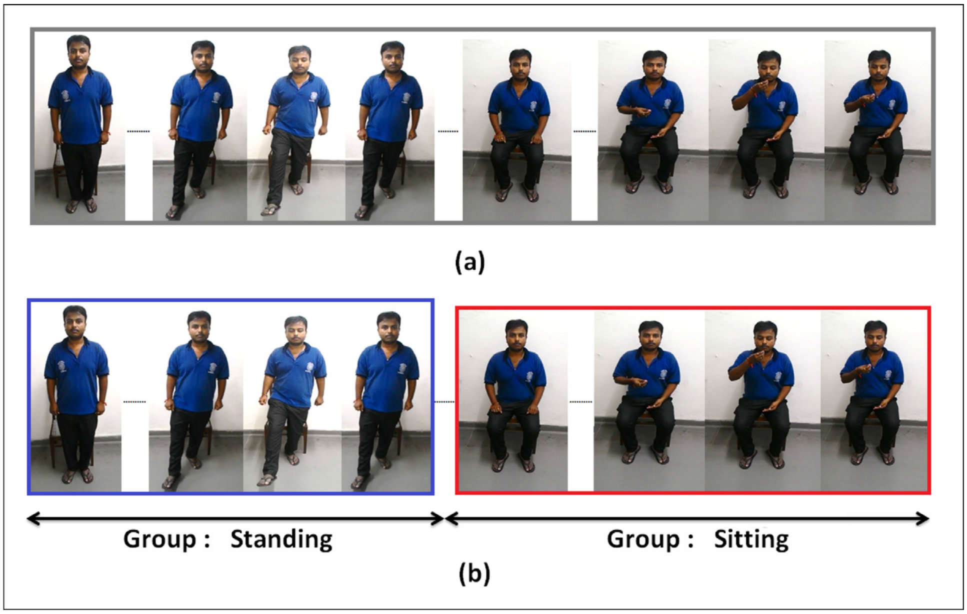

Fig. 6. Example of stage 1: (a) continuous activity sequence (b) grouping of activities as performed during sitting and standing.

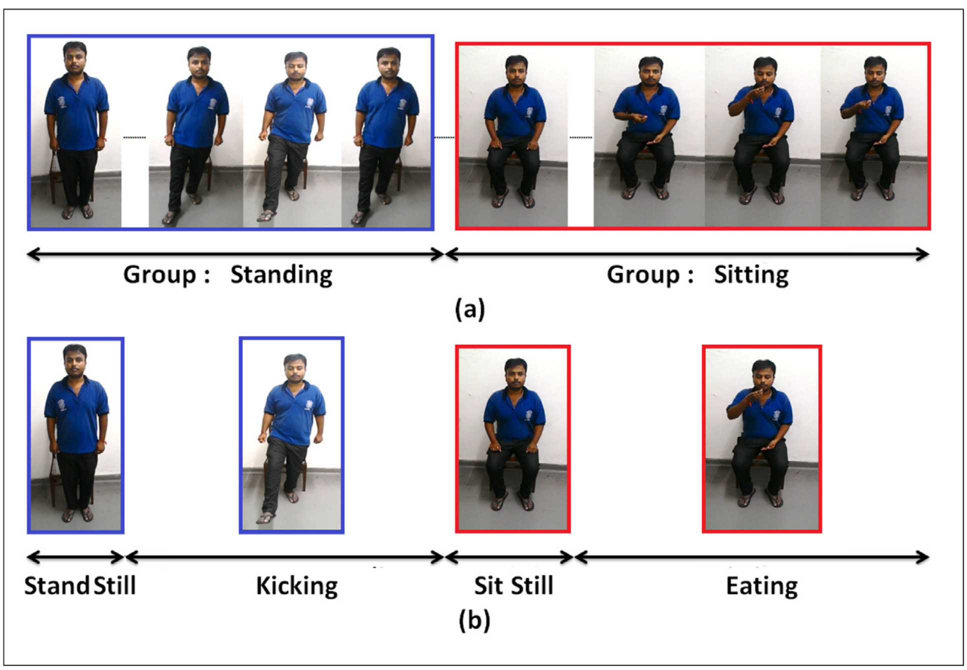

Fig. 7. Example of stage 2: (a) grouped activities (b) recognition of individual activities present in the groups.

BLSTM, etc. However, in this work, after performing multiple experiments, we have adopted BLSTM-NN sequential classifiers based on its performances in activity recognition [10]. BLSTM has been used for recognition of activities in both stages i.e. in stage 1, it is used to recognize sitting and standing activities whereas in stage 2, the classifier recognizes the finer level activities performed during sitting and sitting. To align the activities after coarse-level recognition in stage 1, Viterbi decoder is used for alignment of activities.

3.3.1. Stage 1: Activity recognition and alignment

In stage 1, the given action sequence is first recognized using BLSTM-NN based activity transcription for grouping activities into two groups, where one group consists of activities that are performed during sitting, whereas the other group contains activities performed during standing. Such a group of activities is shown in Table 2, whereas rest of the activities have been performed in both standing and sitting positions. After recognizing the coarse-level groups in the continuous sequence, these groups are segmented. Group alignment using BLSTM-NN is not easy because the output produced by the CTC layer has a probability distribution over activity transcription which does not match directly to the activities present in the input sequences. For this, we have used HMM classifier based forced alignment which is popularly used for consistent and accurate segmentation tasks [45]. In this work, the transcription output of the BLSTM-NN along with the observation sequence is passed to the Viterbi decoder for forced alignment. Using this, the decoder finds an optimal mapping between the observations and the model states that represents the transcription. Each feature vector of the observed sequence is absorbed by one state and similarly each state consumes at-least one feature vector. Once the

Table 2
Grouping of activities performed during standing and sitting.

Activities during standing	Activities during sitting	Activities during standing/sitting
Boxing	Eating	Bending
Dancing	Read sitting	Clapping
Jumping	Sit still	Hand wave
Kicking	Sitting	Phone call
Read standing	Typing	Paper toss
Running	Write sitting	Push/pull
Standing		Thinking
Stand still		Drinking
Walking		
Write standing		

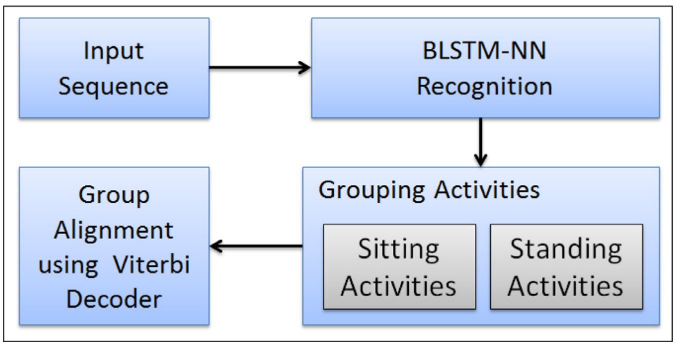

Fig. 8. Flow diagram of the proposed stage 1 processing for grouping and aligning activities.

optimal alignment is done, we find the position and length of the activities using feature vector indices.

Given an observation sequence $x^n = (x_1, x_2, \dots, x_n)$ from a hidden Markov process $X^n = (x_1, x_2, \dots, x_n)$, alignment problems consists of segmentation based on realization of elements of underlying Markov chain $Y^n = (y_1, y_2, \dots, y_n)$ i.e. a mapping g: $\chi^n \rightarrow S^n$ is defined that maps each observation sequence x^n into a sequence of states S^n. An optimal state sequence is achieved by minimizing the risk or in other words maximizing the point wise posterior probability as given in (7). A flow diagram of the approach is shown in Fig. 8.

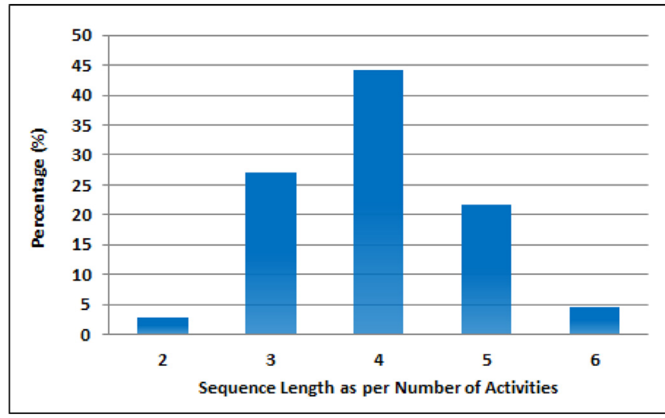

Fig. 9. Distribution of sequence lengths as per the number of activities in dataset. Note: In the dataset the number of activities per sequence varies from 2 to 6 and the height of each bar represent the share of such sequences in the dataset.

$$g(x^n) = arg \max_{s^n} P(Y^n = s^n | X^n = x^n) \tag{7}$$

The state transition probabilities $A = [a_{ij}]$ (transition from state i to state j) and the observation probabilities $B = [b_{jk}]$ (observing symbol k from state j) are computed using forward and backward algorithms and Baum–Welch algorithm [24]. Gaussian Mixture Model (GMM) [25] density has been used for the computation of observation probabilities in the HMM model $\lambda = (A, B, \pi)$ mentioned in [24], where π is the initial probability distribution.

3.4. Stage 2: Recognition of aligned sequences

Stage 2 utilized the output of Stage 1 and process the aligned sequence for the recognition of activities. This has been done using BLSTM-NN classifier. The Connectionist Temporal Classification (CTC) objective function (O) is used as negative *log* probability of correct labeling of the entire training set. CTC has the ability to directly output the probability distribution over a labeled sequence without requiring pre-segmented data. In this work, the classifier

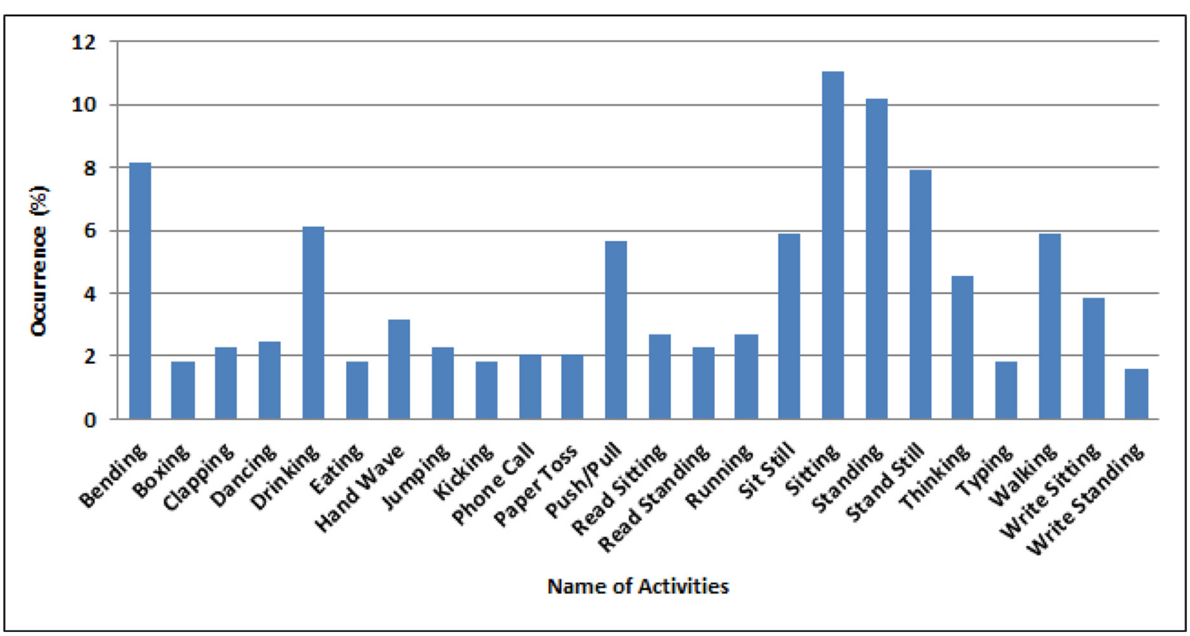

Fig. 10. Percentage share of occurrence of all the 24 activities involved in the dataset.

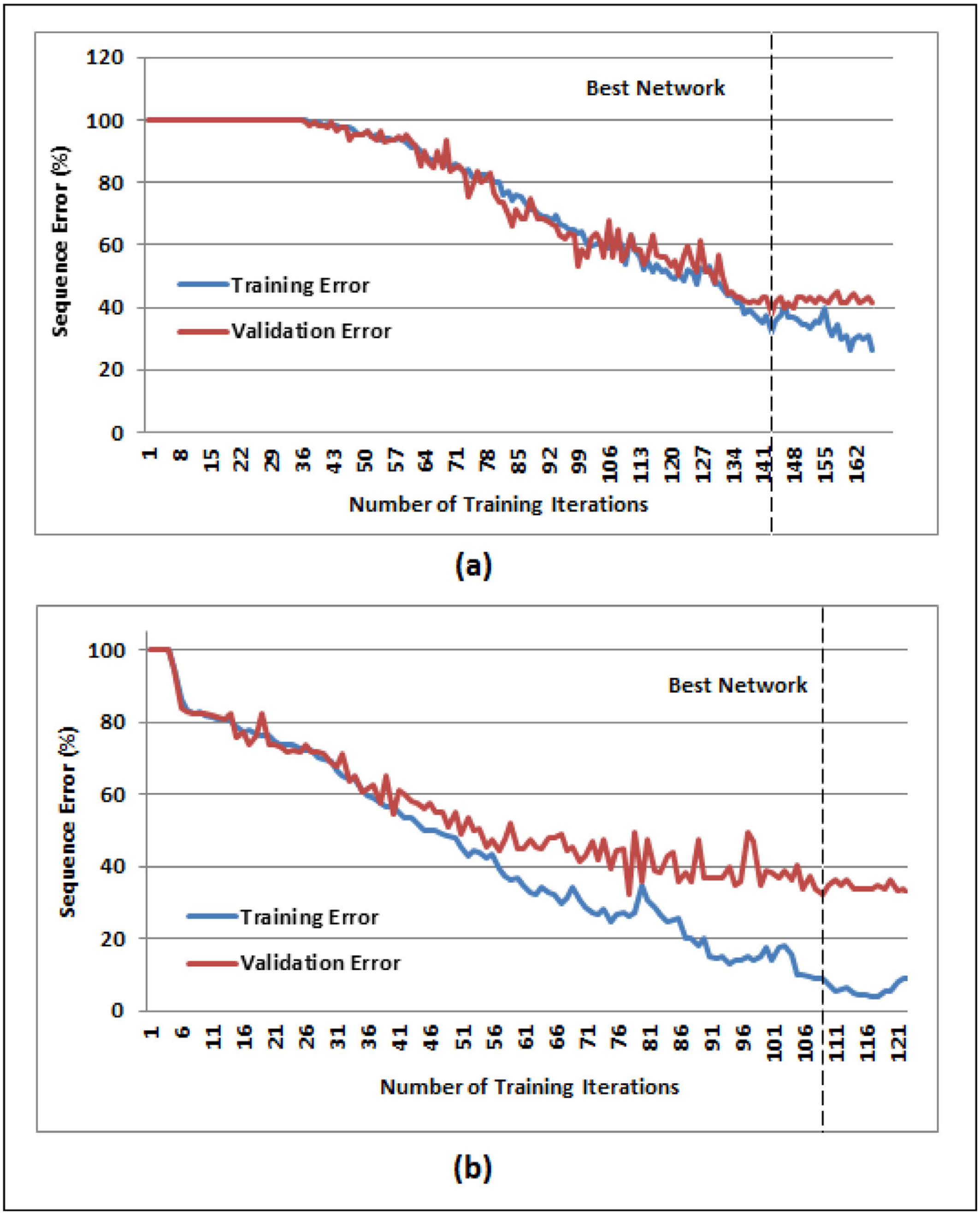

Fig. 11. BLSTM-NN Learning curve showing variation in training and validation errors: (a) Learning curve for complete sequences recognition without sequence alignment (b) Learning curve with sequence alignment.

has been trained with the feature vectors F_T for the recognition of continuous activities.

We have noted that, with Coarse and Fine level stages during activity recognition, performance has improved significantly. It is mainly due to the use of smaller vocabulary in each stage of classification.

4. Results

In this section, we discuss the details of the collected dataset and the computation of activity recognition results. The results have been evaluated using 10-fold cross validation i.e. by keeping 90% of the dataset for training and keeping the rest for testing. The details are as follows.

4.1. Dataset description

Dataset collection was performed using Kinect sensor. We have selected 24 different activities that are commonly performed by persons as regular activities. A total of 111 sequences were prepared using different combinations of activities and performed by 10 different persons (6 males and 4 females). Hence a total of 1110 continuous activity sequences were recorded. The length of the activity sequences in the dataset varies from 2 to 6 number of activities. Fig. 9 shows the distribution of activities as per the length of dataset sequences. It can be seen from the figure that, 44.14% of the sequences contain 4 activities. Fig. 10 shows the percentage of occurrence of all 24 activities involved in this study. The ground truth has been made manually as accurate as possible by carefully analyzing activity sequences for training of the proposed

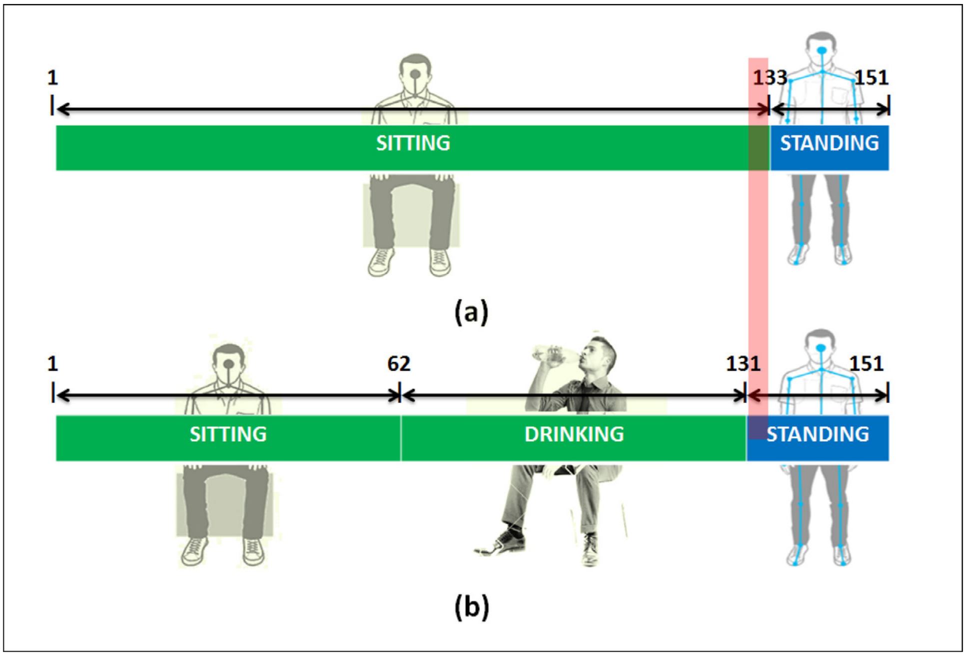

Fig. 12. Correct activity alignment example using Viterbi decoding (a) alignment result of sitting and standing activity (b) corresponding ground truth (GT).

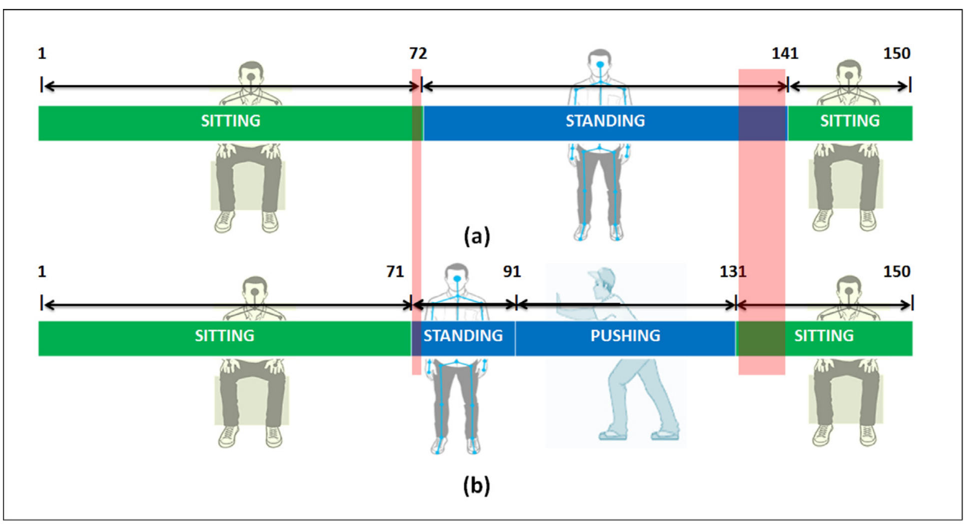

Fig. 13. Erroneous activity alignment example (a) alignment result of sitting-standing-sitting sequence with 12 miss-aligned frames (b) corresponding ground truth of the activity sequence.

system. An activity sequence annotated with ground truth is shown in Fig. 12(b). The dataset has been made online[1] to the research community for further study.

4.2. Activity recognition using conventional approach

Here, we present recognition results of continuous activity sequences using the conventional approaches. In these approaches, classification is performed by training the classifier with the recorded sequence feature vector and testing the given input sequence without doing any pre-segmentation and alignment. For this purpose, BLSTM-NN and HMM based sequence classifiers were used and trained on the feature vectors F_T. The details are as follows.

The BLSTM-NN network is bidirectional and has a hidden layer of LSTM memory blocks to circumvent the information decay. The network has been trained with a learning rate of $1e-4$ and a momentum of 0.9. In each run, the network is initialized with the Gaussian distribution of zero mean and a standard deviation of 0.1. The CTC network has been trained for activity based recognition present in the 3D test sequence without providing dictionary. Fig. 11(a) depicts the learning curve of the BLSTM-NN which shows the error variations in training and validation data. It can be seen from the learning curve, that after 141 training epochs, the validation network becomes constant. Therefore, it is marked as the `Best Network`. The network has been tested for recognizing the activities present in a test sequence. Recognition rate of 64.45% has been recorded with full sequence activity recognition when tested without providing dictionary.

Recognition has also been carried out using HMM classifier which is widely used for modeling temporal sequences [18,23]. The

[1] https://sites.google.com/site/iitrcsepradeep7/.

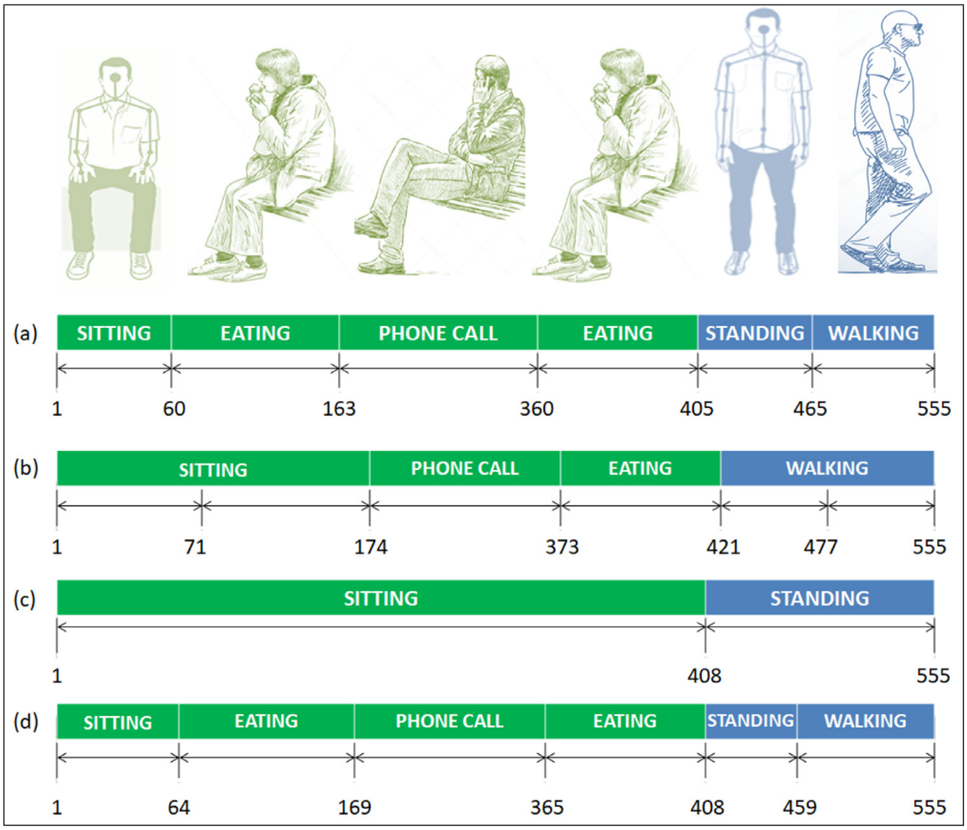

Fig. 14. Activity recognition results: (a) a sequence of activities performed by a person with frame numbers (b) erroneous results when computed using traditional approach of activity recognition (two activities 'sitting' & 'eating' are recognized as 'sitting'. Similarly, 'standing' & 'walking' are recognized as 'walking' only.) (c) Coarse-Level activity recognition (Proposed) (d) Fine-Level activity recognition with correct results (Proposed).

classifier processes an input sequences using left-to-right continuous density HMMs. The basic HMM model (λ) can be defined using $[\pi, A, B]$, where π, A, and B are the initial state, state transition, and observation probability matrices, respectively. A Gaussian Mixture Model (GMM) is defined separately for each state of the model. The recognition accuracy with HMM has been computed by varying number of HMM states $S_t \in \{3, 4, 5, 6\}$ and by varying number of GMM components per state from 1 to 128 with an increment of power of 2. Recognition accuracy of 21.45% has been recorded with 3 HMM states and 2 Gaussian distributions.

4.3. Activity recognition using proposed two stage scheme

In this section, we present the activity recognition performance using the proposed two-stage scheme. Firstly, we provide the results of sequence alignment performance by grouping the activities into two coarser groups. Secondly, the recognition results of these aligned sequences, are presented.

4.3.1. Coarse-level sequence recognition and alignment results

Since before the actual recognition of continuous activities, we have grouped the input sequences into activities that have been performed during sitting and standing. We present the results of sequence transcription using BLSTM-NN and sequence alignment using Viterbi decoding as discussed in Section 3.3.1. The BLSTM network has been trained with the feature vector F_T with a learning rate of $1e-4$ with 0.9 momentum. Recognition rate of 93.81% has been recorded during transcription of sitting and standing activities. Next, the transcription results are passed with the feature vector sequence for length alignment of standing and sitting activities in Viterbi decoding phase. Fig. 12 shows the alignment results of both activities, where Fig. 12(a) depicts the correct alignment of the test sequence consisting of three activities from frame number 1–90 and 91–120 when compared with the corresponding ground truth (GT) as depicted in Fig. 12(b). However, in some cases, the alignment generates erroneous results as depicted in Fig. 13(a), where the test sequence contains a combination of 4 different activities and results into a total of 12 miss-aligned frames as compared to the ground truth shown in Fig. 13(b). Like this, we have considered total 35 continuous sequences to test the robustness of the sequence alignment results and tested against the ground truths. It has been found that approximately 2.1% of the frames in these sequences results into miss-alignment. In addition to this, we have also computed the results using HMM classifier for grouping activities, where we have recorded 83.23% recognition rate on sitting and standing activities using 3 HMM states and 4 number of Gaussian mixture components. Due to less recognition rate as compared to BLSTM, we have considered BLSTM-NN based transcription for finer level classification.

4.3.2. Fine-level activity recognition results

After alignment of input sequences, the segmented sequences are fed to the BLSTM-NN classifier for the recognition of activities. The network has been trained with a learning rate of $1e-4$ and a momentum of 0.9 using the feature vectors F_T. The network has been initialized with the Gaussian distribution of zero mean and a standard deviation of 0.1. The CTC network has been trained for activity based recognition present in the 3D test sequence without providing dictionary. Fig. 11(b) depicts the learning curve of the BLSTM-NN which shows the error variations in the training and validation data. It can be seen from the learning curve, that after 109 training epochs, the validation network becomes

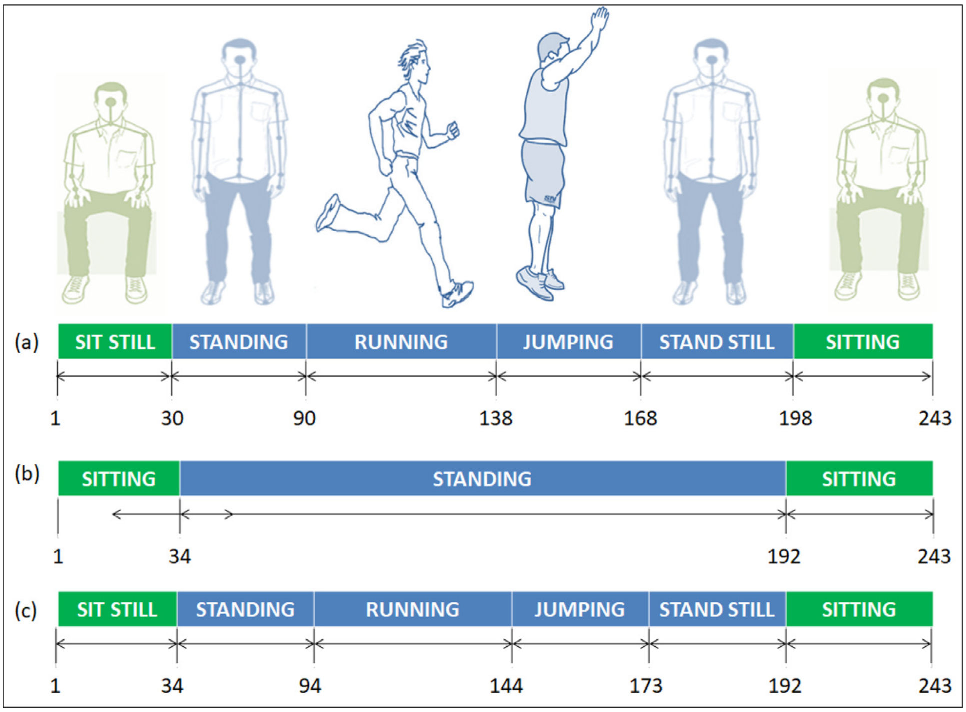

Fig. 15. Activity recognition results: (a) a sequence of activities performed by a person with frame numbers (b) Coarse-Level activity recognition (Proposed) (c) Fine-Level activity recognition with correct results (Proposed).

constant. Therefore, it is marked as the `Best Network`. The network has been tested for recognizing the activities present in a test sequence. Sequence recognition of 68.9% has been recorded when tested without dictionary. The recognition performance is depicted in Fig. 14, where Fig. 14(a) represents a sequence of continuous activities performed by a person along with the number of frames as ground truth (numerical numbers). Fig. 14(b) shows the erroneous recognition results when tested with traditional approach where the approach missed the '*Eating*' & '*Standing*' activities during recognition. Fig. 14(c) and14(d) depict the activity recognition results as per the Coarse-Level and Fine-Level based proposed approach, respectively. It can be seen from the figure that the proposed approach results into correct activity recognition in comparison to traditional approach. Therefore a gain of 4.45% has been recorded in continuous activity recognition results using proposed scheme. Similarly, more results of the proposed scheme in are shown in Fig. 15, where Fig. 15(a) depicts the continuous activities performed a user (tagged with frame numbers per activity), Fig. 15(b) shows the coarse-level division and Fig. 15(c) represents the actual recognition of activities at fine level.

The recognition performance has also been computed as per the number of activities present in test sequences and is shown in Fig. 16, where the sequences with one and four activities depicts the best recognition performance with 82.85% and 73.17%, respectively.

The performance has also been evaluated separately for the activities performed during standing and sitting. Sequence recognition rates of 69.95% and 67.84% have been recorded in activities performed during standing and sitting, respectively. A comparative analysis of the recognition performances in the form of bar chart is shown in Fig. 17.

4.4. Isolated activity recognition results

Moreover, the performance of the system has also been evaluated by recognizing isolated activities. For this, we have aligned

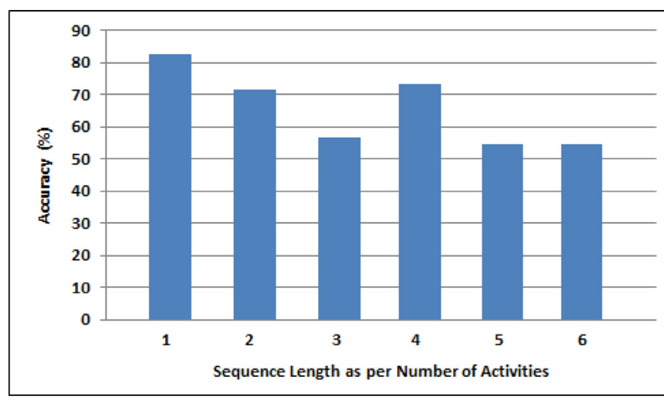

Fig. 16. Activity recognition performance as per the number of activities present in the test sequences.

all the trajectories using Viterbi decoding and then fed them to the sequential classifiers. With BLSTM-NN, an accuracy of 81.9% has been recorded for recognizing isolated activities. The confusion matrix for the isolated activities is shown in Fig. 18. The recognition of individual activities is also performed by HMM [2]. We have obtained 75.23% accuracy using HMM.

4.5. Comparative analysis

Since, there exist no public dataset we could not compare our performance directly. However, to get an idea of continuous activity recognition, we have compared our proposed scheme with some existing state of the art techniques. Two methods have been selected for comparison. Bloom et al. [3] have proposed a continuous HAR recognition for gaming applications consisting of 20 different gaming activities. Second work is presented in [44] for recognition of 16 daily use basic activities recorded using 3D skeletal by Kinect. The comparison is shown in the Table 3, where

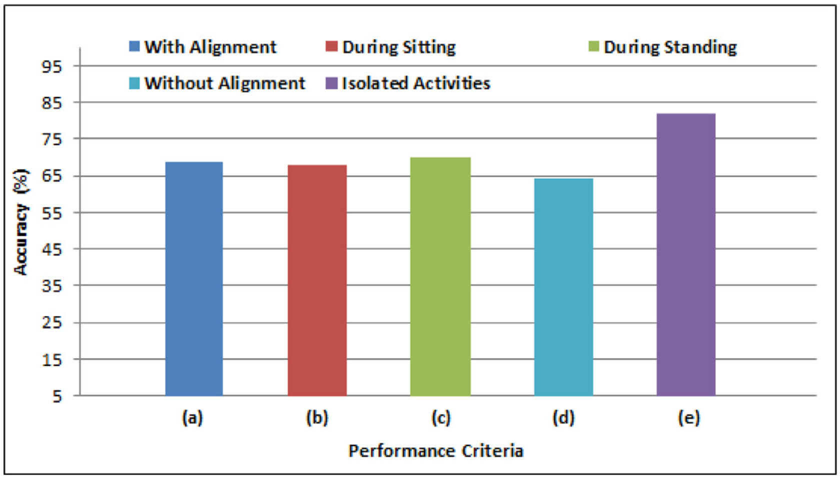

Fig. 17. Comparative performance analysis among different test scenarios: (a) continuous activity recognition results using proposed scheme (b) recognition of activities performed during sitting (c) recognition of activities performed during standing (d) continuous activity recognition results using traditional approach (e) isolated activity recognition performance.

Table 3
Comparative performance analysis with state of the art techniques.

Author name & year	Methodology (Features & classifier)	Dataset	
Bloom et al. [3], 2012	3D joints position, velocity, magnitude, angle velocity	20 gaming activities	61.17%
Zhu et al. [44], 2016	3D joints position, key poses, atomic motions, Maximal entropy Markov model	16 activities	54.7%
Yu et al. [41], 2016	3D joints, pair-wise joint difference, depth, SVM	7 activities	56.4%
Proposed approach	**3D joints, angular, distance and velocity features, grouping activities and segmentation**	**24 activities**	**68.9%**

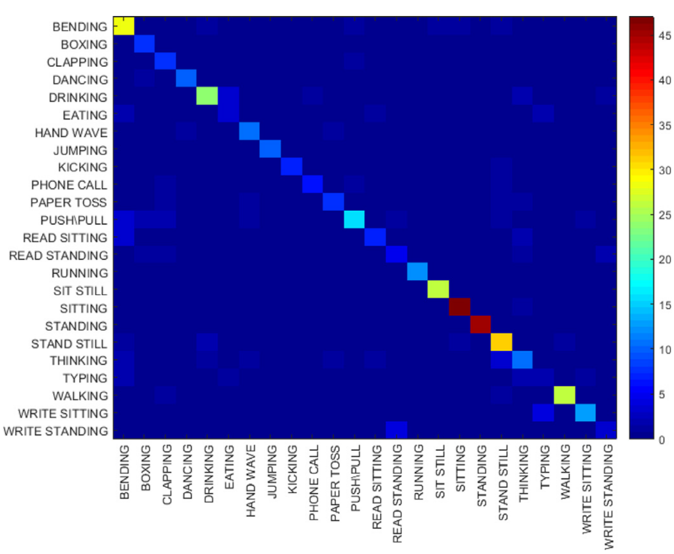

Fig. 18. Confusion matrix for the recognition of isolated activities.

the proposed approach performs better in activity recognition with larger number of activities.

4.6. Error analysis

We have analyzed the errors incurred during activity recognition. The errors mainly have occurred due to two reasons. Firstly, the miss-alignment of activities could be the main reason of erroneous recognition as sometimes it segments the frame of a useful activity into two parts that results into missing information during recognition process and creates a confusion that leads to wrong activity recognition for the entire sequence. Secondly, it has been observed that some users performs certain activity very fast that lead to recording of less number of frames. This creates confusion during the continuous activity recognition process. For example, two activities, namely '*toss paper*' and '*kicking*' are performed in less than a second time duration, therefore, with lesser recorded frames the activities could not be recognized accurately.

5. Conclusion and future scope

In this paper, we have proposed a novel framework for continuous-HAR using Kinect. The continuous activities have been recorded in 3D skeleton form of the human torso. Using this, we have recorded a large dataset of 1110 sequences comprised of 24 isolated activities. To recognize the activities, the recorded sequences are first grouped into activities performed during sitting and standing. Next, these groups are aligned using length modeling performed by Viterbi decoding. Finally, BLSTM-NN has been trained to recognize the continuous activities present in each of these aligned sequences. A sequence recognition rate of 68.9% has been recorded during the performance evaluation of the proposed system. In future, the work can be extended by combining sequence classifiers. Various alternatives for sequence alignment and other HMM models along with different topologies of Neural Network can also be used for better recognition performances.

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